

EUCAST phenotypic screening tests to detect / exclude resistance of clinical relevance

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EUCAST recommends the use of screening agents and disks to detect or exclude resistance mechanisms. Below are screening tests referred to in the EUCAST breakpoint table. Criteria (MIC, zone diameter etc) for negative and positive tests are listed in the breakpoint table.

A general comment on “screening tests” is available in Breakpoint Tables Notes: A screening test uses one agent to detect or exclude resistance or predict susceptibility to one or more antimicrobial agents in the same class. The screening test is often more sensitive and/or robust than testing individual agents. Using a screening test will reduce the number of tests needed in primary susceptibility testing. It is especially useful when resistance is infrequent. Guidance on how to act on the screening test result is in the Note related to each specific test in the breakpoint tables. Recommendations on methods to detect resistance mechanisms for epidemiological purposes are in the EUCAST Guidance document “EUCAST guidelines for detection of resistance mechanisms and specific resistances of clinical and/or epidemiological importance” (https://www.eucast.org/resistance_mechanisms).

Negative screening test: MIC is less than or equal to or zone diameter more than or equal to the susceptible breakpoint for the screening agent. No resistance mechanisms to the antimicrobial class detected. Do not report the screening test as such – only the conclusion of a negative screening test (for example “ciprofloxacin susceptible”, or “the isolate is devoid of any resistance mechanisms to beta-lactam agents” etc).

Positive screening test: MIC is above or zone diameter below the resistant breakpoint for the screening agent. Resistance mechanisms to the antimicrobial class detected. Do not report the screening test as such – only the conclusion of a positive screening test (for example “the *S. aureus* isolate is methicillin-resistant (MRSA)”, “the isolate is resistant to erythromycin and hence also to other macrolides” etc).

EUCAST recommended phenotypic screening tests.

Screening test Antimicrobial, disk	Organism(s)	Resistance to agent(s)	Rationale
Oxacillin 1 µg	<i>Streptococcus pneumoniae</i>	Beta-lactam	Negative screen excludes beta-lactam resistance in <i>S. pneumoniae</i> . Positive screen is because of one or several beta-lactam resistance mechanisms – see flowchart in the <i>S. pneumoniae</i> table of the breakpoint table.
Oxacillin 1 µg	<i>Staphylococcus pseudintermedius</i> , <i>S. schleiferi</i> , <i>S. coagulans</i>	Beta-lactam	Negative screen excludes methicillin resistance. Positive screen may be due to the presence of <i>mec</i> genes coding for

			“methicillin”-resistance or high production of staphylococcal beta-lactamase. Report resistant to beta-lactam agents.
Cefoxitin 30 µg	Staphylococci other than <i>Staphylococcus pseudintermedius</i> <i>S. schleiferi</i> , <i>S. coagulans</i>	Beta-lactam	Negative screen excludes methicillin resistance. Positive screen may be due to the presence of <i>mec</i> genes coding for “methicillin”-resistance or high production of staphylococcal beta-lactamase. Report resistant to beta-lactam agents. Ceftobiprole and ceftaroline may be active against MRSA - test separately.
Benzylpenicillin (PCG 1U)	<i>Haemophilus influenzae</i>	Beta-lactam	Negative screen excludes beta-lactam resistance in <i>H. influenzae</i> . For beta-lactam agents relevant for <i>H. influenzae</i> no further testing is necessary. Positive screen is either because of beta-lactamase production or mutations in penicillin-binding proteins – see flowchart in the <i>H. influenzae</i> table of the breakpoint table.
Benzylpenicillin (MIC or PCG 1U)	Viridans group streptococci	Beta-lactam	Negative screen excludes beta-lactam resistance in viridans group streptococci. Positive screen is because of one or several beta-lactam resistance mechanisms. Test the relevant beta-lactam agent and report according to its specific breakpoints.
Nalidixic acid 30 µg	<i>Haemophilus influenzae</i> <i>Moraxella catarrhalis</i> <i>Pasteurella multocida</i>	Fluoroquinolone	Negative screen excludes quinolone resistance in <i>H. influenzae</i> , <i>P. multocida</i> and <i>M. catarrhalis</i> . Positive screen is because of the presence of one or several quinolone resistance mechanisms. Report resistant to fluoroquinolones or test the relevant quinolone and report according to its specific breakpoints.
Pefloxacin 5 µg	<i>Salmonella</i> spp. <i>Vibrio</i> spp.	Fluoroquinolone	Negative screen excludes fluoroquinolone resistance in <i>Salmonella</i> spp. and <i>Vibrio</i> spp. Positive screen is because of the presence of one or several fluoroquinolone resistance mechanisms. Report resistant to fluoroquinolones or test the relevant fluoroquinolone and report according to its specific breakpoints.
Norfloxacin 10 µg	<i>Staphylococcus</i> spp. <i>Streptococcus</i> ABCG <i>Streptococcus pneumoniae</i> <i>Enterococcus</i> spp., <i>Bacillus</i> spp., <i>Aerococcus</i> spp.	Fluoroquinolone	Negative screen excludes fluoroquinolone resistance in streptococci, enterococci and staphylococci. Positive screen is because of the presence of one or several fluoroquinolone resistance mechanisms. Report resistant to fluoroquinolones or test the relevant fluoroquinolone and report according to its specific breakpoints.
Erythromycin 15 µg	<i>Staphylococcus</i> spp. <i>Streptococcus</i> ABCG <i>S. pneumoniae</i> <i>Campylobacter</i> spp. <i>Vibrio</i> spp.	Azithromycin, clarithromycin and roxithromycin	Negative screen excludes resistance to azithromycin, clarithromycin and roxithromycin and these can be reported susceptible without further testing. Positive screen is because of macrolide resistance. Report resistant to macrolides or test the relevant macrolide and report according to its specific breakpoints.
Tetracycline 30 µg	<i>Staphylococcus</i> spp. <i>Streptococcus</i> ABCG <i>Streptococcus pneumoniae</i>	Doxycycline and minocycline	Negative screen excludes resistance to doxycycline and minocycline and these can be reported susceptible without further testing.

	<i>Haemophilus influenzae</i> <i>Moraxella catarrhalis</i> <i>Pasteurella multocida</i> <i>Campylobacter jejuni</i> and <i>C. coli</i> <i>Kingella kingae</i> <i>Vibrio</i> spp. <i>Burkholderia pseudomallei</i>		Positive screen is because of resistance to tetracyclines. Report resistant or test relevant tetracycline and report according to its specific breakpoints.
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